Task 5 Numerical Methods

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PROBLEM 1

I use the Hungarian Algorithm to solve the assignment problem. The initial cost matrix is:

$$\begin{pmatrix} 3 & 3 & 1 & 2 & 10 \\ 1 & 9 & 9 & 4 & 10 \\ 9 & 9 & 3 & 10 & 9 \\ 9 & 10 & 1 & 7 & 6 \\ 10 & 7 & 7 & 1 & 3 \end{pmatrix}$$

STEP 1: ROW AND COLUMN REDUCTION

Row Reduction

Subtract the smallest element in each row from every element in that row. The matrix becomes:

$$\begin{pmatrix} 2 & 2 & 0 & 1 & 9 \\ 0 & 8 & 8 & 3 & 9 \\ 6 & 6 & 0 & 7 & 6 \\ 8 & 9 & 0 & 6 & 5 \\ 9 & 6 & 6 & 0 & 2 \end{pmatrix}$$

Column Reduction

Subtract the smallest element in each column from every element in that column. The matrix remains the same in this case:

$$\begin{pmatrix} 2 & 0 & 0 & 1 & 7 \\ 0 & 6 & 8 & 3 & 7 \\ 6 & 4 & 0 & 7 & 4 \\ 8 & 7 & 0 & 6 & 3 \\ 9 & 4 & 6 & 0 & 0 \end{pmatrix}$$

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STEP 2: COVERING ZEROS WITH THE MINIMUM NUMBER OF LINES

Draw lines to cover all the zeros in the matrix. I can cover all zeros with 3 lines (vertical):

$$\begin{pmatrix} 2 & [0] & [0] & 1 & 7 \\ 0 & 6 & 8 & 3 & 7 \\ 6 & 4 & [0] & 7 & 4 \\ 8 & 7 & [0] & 6 & 3 \\ 9 & 4 & 6 & [0] & [0] \end{pmatrix}$$

STEP 3: ADJUSTING THE MATRIX

Find the smallest non-covered element, which is 1. Subtract 1 from all non-covered elements to get:

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 6 \\ 0 & 5 & 7 & 2 & 6 \\ 5 & 3 & 0 & 6 & 3 \\ 7 & 6 & 0 & 5 & 2 \\ 8 & 3 & 5 & 0 & 0 \end{pmatrix}$$

STEP 4: COVERING ZEROES AGAIN

Cover all zeros with 5 lines (horizontal), one for each row:

$$\begin{pmatrix} [1] & [0] & [0] & [0] & [6] \\ [0] & [5] & [7] & [2] & [6] \\ [5] & [3] & [0] & [6] & [3] \\ [7] & [6] & [0] & [5] & [2] \\ [8] & [3] & [5] & [0] & [0] \end{pmatrix}$$

STEP 5: MAKING ASSIGNMENTS

One optimal assignment is as follows:

- Assign the first worker to the second job (0 at [1, 2])
- Assign the second worker to the first job (0 at [2, 1])
- Assign the third worker to the third job (0 at [3, 3])
- Assign the fourth worker to the fourth job (0 at [4, 4])
- Assign the fifth worker to the fifth job (0 at [5, 5])

The total cost of this assignment is 17.

PROBLEM 2